Agentic Engineering at Regulated Scale: Building a Post-Quantum, FHE-Native Securities Platform

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Abstract

We document the engineering capability behind Hanzo's stack by way of its hardest production proof: an agent-driven team built a fully regulated securities venue—an SEC-registered alternative trading system (ATS) with broker-dealer and transfer-agent functions—on a next-generation foundation of production post-quantum cryptography, a high-performance native-GPU blockchain, and fully-homomorphic-encryption (FHE) privacy. The build *consolidated* rather than sprawled—200+ microservices collapsed into 3 compiled binaries, with 3.07M lines of code reduced to 1.06M active (∼3.5M lines of dead code deleted)—reaching production in record time with a small team, and passed FINRA/SEC approval and SOC 2 compliance. This is not template web software; it is among the most compliance-heavy and cryptographically advanced software domains that exists. We describe the throughput of the agentic engineering method that produced it, and we expand on the defense relevance of its FHE layer—computing on data one is not permitted to see.

Executive Summary. The fastest way to judge an engineering organization is what it has shipped under real constraints. Hanzo's founding CTO, Zach Kelling, was ranked the #**1 user of Anthropic by token count worldwide in 2025** (per public usage tracking at `claudecount.com`)—a marker of an engineer operating agentic AI at the frontier of throughput. That throughput is not spent on brochure sites: the flagship production system built with Hanzo's agentic stack is a **fully regulated securities exchange**—an SEC-registered ATS with broker-dealer and transfer-agent functions—running on **post-quantum cryptography**, a **native-GPU high-performance blockchain**, and **fully homomorphic encryption** for privacy. The build *consolidated* rather than sprawled—200+ microservices collapsed into 3 compiled binaries, 3.07M lines of code cut to 1.06M active—shipped fast with a small team, and cleared FINRA/SEC approval and SOC 2. Regulated financial infrastructure is one of the hardest, most audited software domains there is; clearing it on a cryptographically advanced foundation is the proof that agent-driven engineering is production-grade, not a demo—and that it is exactly the capability needed to modernize defense software stacks.

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1 The Engineer and the Throughput

Agentic engineering multiplies a strong engineer; it does not replace one. Hanzo's founding CTO works at the frontier of both AI/ML and cryptography—the two disciplines this stack fuses—and operates agentic AI at a volume that placed him, by public token-count tracking (`claudecount.com`), as the top Anthropic user in the world in 2025. The relevant point is not the ranking for its own sake; it is what that volume was spent producing. Raw AI usage that yields disposable prototypes proves nothing. Raw AI usage that yields a regulated, audited, cryptographically advanced production system proves the method works where it is hardest.

2 What Was Built

The flagship system is a regulated securities venue, not a content site. In U.S. securities terms it spans three regulated functions that ordinarily take separate firms years to stand up:

- **Alternative Trading System (ATS)** — an SEC-registered trading venue (Regulation ATS), matching and executing orders.
- **Broker-Dealer (BD)** — FINRA-regulated, with the supervisory, recordkeeping, and capital obligations that entails.
- **Transfer Agent (TA)** — SEC-registered, maintaining the authoritative record of securities ownership and transfers.

These functions are governed by some of the most exacting compliance regimes in software. The notable engineering fact is that the build *consolidated* rather than sprawled: a legacy estate of **200+ microservices** on a public cloud was collapsed into **3 compiled binaries**, and the codebase went from **3.07M lines to 1.06M active** (~70% smaller, with ~3.5M lines of dead code deleted; the trading/auth core itself shrank from ~264,000 to ~44,000 lines, with 1.3 GB of vendored dependencies eliminated and a single 48 MB binary subsuming 17+ formerly separate services). It reached production **in record time**, with a **small team**, and was carried through **FINRA/SEC approval** and **SOC 2** compliance—leaner than the system it replaced, not larger.

3 One Engineer, a Team's Output

The result that matters most is the ratio of output to headcount. The agentic method let a *single* AI-augmented engineer carry the bulk of a regulated production platform that a conventional organization staffs as a full team—and the project's own git history documents it. In a **10-day remediation sprint**, one engineer produced **3,694 commits**, added ~55,000 lines and removed **~3.5 million lines of dead code** across **181 repositories**, and authored **100% of the critical- and high-severity security fixes**—single-handedly—while driving the consolidation of 200+ microservices into three binaries. The security and compliance gaps were closed on the path to approval (a source audit of **93 findings**, 22 critical, plus **780+ dependency vulnerabilities** remediated to zero actionable), through FINRA/SEC approval and SOC 2.

The economic consequence is direct. When one engineer, multiplied by the agentic stack, produces what a team produces, the **cost per line of shipped, audited, production code** falls by a large factor relative to a conventional staffing contract. For a defense or government program the implication is the whole argument: the same regulated, post-quantum, FHE-native system, delivered by a fraction of the headcount at a fraction of the cost—and owned, not rented. This is what "agentic engineering" means in practice—not a chatbot writing snippets, but one expert operating at the throughput of a department.

4 The Foundation: Post-Quantum, GPU Blockchain, FHE

What makes the build a genuine engineering flex is the foundation it runs on— each element of which is independently hard:

- **Production post-quantum cryptography.** All three NIST PQC standards (ML-KEM/203, ML-DSA/204, SLH-DSA/205) in a live system with hybrid classical/quantum security—so the chain of custody for securities records is protected against a future quantum adversary.
- **Native-GPU high-performance blockchain.** A chain serving as the authoritative, immutable, post-quantum-anchored record of transfers, with GPU-accelerated cryptographic operations for throughput.
- **Fully homomorphic encryption (FHE).** GPU-accelerated FHE (BFV/BGV/CKKS/TFHE) with threshold decryption, so privacy-sensitive computation runs *on ciphertext*—no single component ever holds both the plaintext and the result.

A team that can compose post-quantum cryptography, a GPU blockchain, and FHE into a system that clears securities regulators is a team that can modernize a defense software stack to the same foundation.

5 FHE for the Military: Computing on Data You Cannot See

Fully homomorphic encryption is the capability with the sharpest defense upside, so it is worth drawing out. FHE allows arbitrary computation directly on encrypted data: the operator performs the computation and returns an encrypted result without ever decrypting the inputs, and only the data owner (or a threshold of keyholders) can decrypt the output. The system doing the work never sees the plaintext. In the securities platform this protects order and ownership data; the same primitive answers several defense problems that have no clean classical solution:

- **Cross-domain transfer on ciphertext.** A cross-domain guard can evaluate a release policy against encrypted content and decide whether data may move between security levels *without ever seeing the data*—removing the guard itself as a point of compromise (developed in the cross-domain paper).
- **Coalition analytics without disclosure.** Allies can compute a joint result—shared targets, overlapping indicators, combined logistics—over their combined data while each party's raw data stays encrypted and private. Nobody has to surrender sources or methods to get the joint answer.
- **Private intelligence fusion.** Multi-source correlation can run while the sources remain encrypted, so a breach of the fusion system does not expose the underlying collection.
- **Encrypted ML inference.** A model can classify encrypted imagery or signals: the data owner never exposes the input, and the model operator never sees it—useful when the data is sensitive, the model is sensitive, or both.

- **Computation on untrusted infrastructure.** Sensitive workloads can run on contractor or shared infrastructure without exposing the data to the host.

FHE's historical objection is speed; the practical answer in this stack is GPU acceleration of the underlying number-theoretic transforms, plus threshold decryption so no single party holds the key. "Compute on data you are not allowed to see" is a capability defense has wanted for decades; here it is a production component, not a research slide.

6 Why Open Source Was the Accelerant

The obvious question is how a small team shipped a regulated, cryptographically advanced system of this scale so quickly. The answer is open source—specifically, an *owned* open-source stack composed by an agentic engineering method.

Hanzo did not assemble the platform from a patchwork of third-party SaaS. It composed it from coherent open-source building blocks that Hanzo itself authors and controls—identity and access management, key management, the data layer, the gateway and ingress, the ZAP transport, the post-quantum and FHE cryptography, the operator, and the Zen models—each reusable, inspectable, and owned. The agentic stack is the multiplier; the owned OSS foundation is the leverage. You do not rebuild identity, key management, or a gateway for each new system; you compose owned, hardened components and point the agents at the application on top. That is how a regulated platform of this scope reaches production fast—and lean.

For defense, open source inverts the usual security argument. A closed system asks you to trust a vendor you cannot inspect; an open one lets you read every line, verify it—here, with machine-checked proofs—confirm there are no hidden backdoors, and run it on hardware you control. Auditable beats opaque whenever the stakes are high. Open source is therefore not a cost-cutting compromise: it is simultaneously the source of **velocity** (compose, do not rebuild), of **security** (inspectable and formally verifiable), and of **sovereignty** (own it, modify it, run it anywhere, including air-gapped). The three properties a defense program most wants are the same three that made the build fast.

Stated plainly as a hook: *Hanzo builds massive, secure systems quickly because it builds on an open-source stack it owns—so the speed, the security, and the sovereignty all come from the same source.*

7 Why It Matters for Defense Modernization

The throughput and the difficulty compound. The same agentic engineering method that shipped a regulated, post-quantum, FHE-native securities exchange fast and small is the method Hanzo brings to defense and government modernization: take an existing, already-accredited stack and modernize it in place—to a sovereign, post-quantum, self-hostable footprint with FHE-grade privacy—faster and cheaper than a traditional rebuild, and owned by the operator. The securities venue is the existence proof that the method holds up under the most demanding regulatory and cryptographic constraints civilian software offers; defense modernization is the same engine pointed at the mission.